



PARALA MAHARAJA ENGINEERING COLLEGE

Summer Internship on Research sensitization & innovative project development

Topic: Multimodal Explainable AI & Mental Health

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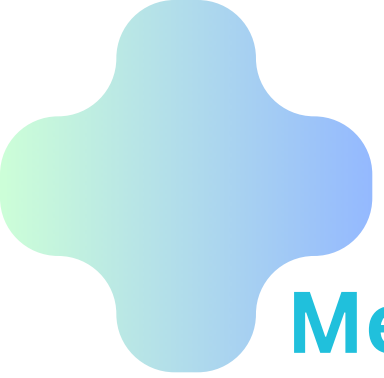
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CONTENTS

1. Introduction
2. Domain & Subdomain
3. Problem Description
4. Literature Survey & Research Gaps
5. Problem Statement
6. Objectives
7. Functional & Non-Functional Requirements
8. Technology Stack
9. System Architecture
10. UML Diagrams (Use Case, Class & Sequence)
11. Implementation & Working
12. Results & Explainable AI Analysis
13. Web Application Demonstration
14. Conclusion & Future Scope
15. References



Introduction

Mental Health

- According to the **World Health Organization**, nearly one in eight individuals worldwide lives with a mental health condition, yet a significant proportion remains undiagnosed.
- Some major mental disorders are: **Depression, Anxiety, PTSD, Dysthymia, Bipolar I Disorder, Bipolar II Disorder etc.**
- The rise of **Artificial Intelligence** has provided tools for enabling personalized experiences and prediction of disorder, intelligent self-assessment, real-time emotional support, and data-driven insights that adapt to each individual's unique needs.

Influence of Family and Loneliness Risk On Mental Health

- Family environment and social connectedness are major psychosocial factors that significantly influence an individual's mental health and emotional well-being.
- Supportive family relationships and strong social bonds act as protective factors, reducing the risk of depression, anxiety, stress, and other mental health disorders.
- Dysfunctional family dynamics, frequent conflicts, financial stress, and lack of emotional support can increase vulnerability to mental health problems and delay recovery.
- Persistent loneliness and social isolation are associated with higher risks of depression, anxiety, psychological distress, suicidal ideation, and reduced quality of life.
- Understanding family influence and loneliness alongside clinical symptoms enables more accurate mental health risk assessment and supports personalized intervention and treatment planning.

Domain & Subdomain

Major Domain (Broader Area)

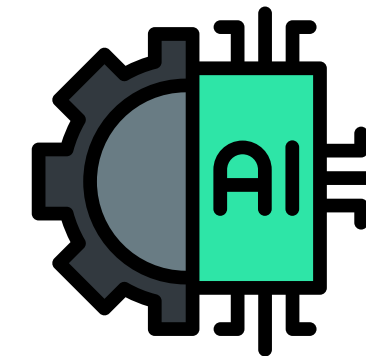
Mental Health

Study and assessment of psychological well-being, disorders, and early intervention strategies



Artificial Intelligence & Machine Learning (AI/ML)

Application of intelligent algorithms to learn from data and make predictions.



Subdomain (Narrower Area)

- **Mental Health Prediction & Screening:** Use of computational models to identify early signs of mental health conditions through structured assessments.
- **Explainable AI (XAI):** A branch of AI focused on making model predictions transparent, interpretable, and trustworthy for end users and professionals.
- **Multimodal Machine Learning:** Integration of multiple types of input data (such as text, questionnaire responses, behavioral patterns) to improve prediction accuracy.



Literature Review and Research Gaps



Article 1: Explainable AI for Mental Health Diagnosis [3]

Given/Written By	UK NHS-based Explainable AI for Mental Health Diagnosis Study
Dataset Used	500 real NHS patient cases (Adult ADHD)
Model Used	Hybrid Model (ML Binary Classifier + Knowledge-Based Rule System)
Brief Description	Combines machine learning and expert-defined clinical rules for transparent mental health diagnosis; clinician review for conflicting predictions.
Accuracy	98% Accuracy with explainable decision-making.
Research Gaps	No standard explanation metric; symptom overlap issues; rule-based systems have poor scalability.

Article 2: Toward Explainable AI (XAI) for Mental Health Detection Based on Language Behavior [2]

Given/Written By	Elma Kerz, Sourabh Zanwar, Yu Qiao, and Daniel Wiechmann
Dataset Used	Reddit, Twitter, online forums, and clinical text datasets
Model Used	SVM, Random Forest, Decision Tree, Logistic Regression, Naïve Bayes, BERT with SHAP & LIME
Brief Description	Uses NLP, Deep Learning, and XAI to detect depression and anxiety while explaining influential words and sentences.
Accuracy	Performance compared using ML/DL metrics; XAI improves interpretability (specific accuracy not reported).
Research Gaps	limited privacy-sensitive datasets; need for multilingual and cross-cultural validation.

Article 3: Multimodal Machine Learning in Mental Health — A Survey of Data, Algorithms, and Challenges [9]

Given/Written By	Alsahili, Patras & Purver (Queen Mary University of London)
Dataset Used	26 public multimodal mental health datasets
Model Used	Hybrid CNN/RNN, Transformer-based Models, Graph Neural Networks (GNNs)
Brief Description	Survey on multimodal AI using speech, text, facial, and physiological data for mental health screening.
Accuracy	Transformer: F1 = 0.967 ; GNN: F1 = 0.95
Research Gaps	Small biased datasets, lack of benchmarks, black-box models, fairness issues, limited handling of comorbidity.

Article 4: Explainable AI for Mental Health through Transparency and Interpretability for Understandability (TIFU Framework) [1]

Given/Written By	Dan W. Joyce, Andrey Kormilitzin, Katharine A. Smith & Andrea Cipriani
Dataset Used	PubMed review of 15 XAI mental health studies (2018–2022)
Model Used	TIFU Framework; Logistic Regression (reference model); SHAP & LIME discussed
Brief Description	Defines transparency and interpretability standards for explainable AI in mental healthcare.
Accuracy	No predictive accuracy reported; only 3 studies tested clinician understanding.
Research Gaps	No clear explainability definitions, limited clinician validation, SHAP/LIME alone do not ensure true interpretability.

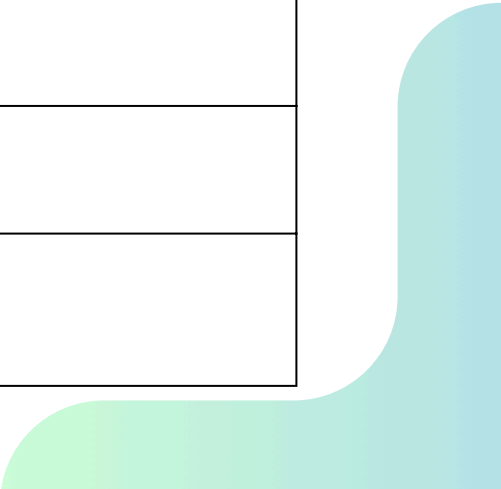


Article 5: A multimodal fairness aware machine learning framework for mental health risk prediction in university students [10]

Given/Written By	Zhu Tian, Jian Zhang & Xia Wang
Dataset Used	14,604 students from 24 universities
Model Used	BD-MHAM (Multimodal Fairness-Aware ML Framework
Brief Description	Predicts student mental health risk using multimodal data with SHAP-based explainability.
Accuracy	AUC = 0.972, Accuracy = 95.6%, F1 = 0.928
Research Gaps	Small datasets, lack of multimodal integration.

Article 6: An explainable multimodal deep learning approach for stress detection in emotion-aware systems [11]

Given/Written By	Sai H. Prajwal, Sofia Singh & Dipti Theng
Dataset Used	WESAD dataset (15 participants)
Model Used	Random Forest + SHAP + Wavelet-based HRV Features
Brief Description	Explainable multimodal framework for stress detection using physiological signals.
Accuracy	Accuracy = 83%
Research Gaps	Small dataset, limited modalities, need for larger validation and improved interpretability.



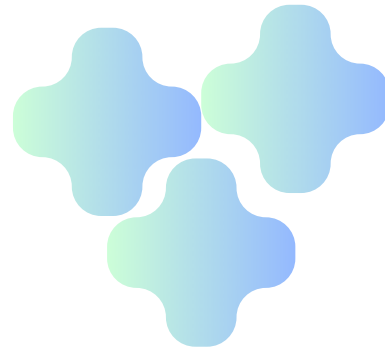


Article 7: Explainable artificial intelligence approaches for predicting depression by combining feature selection methods and machine learning classifiers [4]

1. Given/Written By	Min Gyeong Kim, Kun Chang Lee, Kwanho Lee, Hyung Uk Kim, Young Wook Seo & Seong Wook Chae
2. Dataset Used	National Mental Health Survey of Korea (2021) containing 5511 adults .
3. Model Used	Feature Selection Methods (ReliefF, Markov Blanket, CFS) with ML Classifiers and SHAP-based XAI.
4. Brief Description	Developed an explainable AI framework for depression prediction and identified important risk factors such as loneliness, social distress, and quality of life indicators.
Accuracy/Metrics	Best performance achieved F2-Score = 98.51% using ReliefF with Stacking classifier.
6. Research Gaps	Limited cross-cultural validation, potential circular reasoning in prior studies, and lack of systematic comparison between feature selection methods and classifiers.

Article 8: Integrating Large Language Models into a Tri-Modal Architecture for Automated Depression Classification [5]

1. Given/Written By	Santosh V. Patapati
2. Dataset Used	DAIC-WOZ – 189 clinical interviews with audio, transcripts, facial features (FAUs), and PHQ-8 depression labels.
3. Model Used	Tri-Modal BiLSTM using MFCC + FAUs + GPT-4 with Model-Level Fusion.
4. Brief Description	Combines audio, video, and text features to classify Major Depressive Disorder (MDD) using a BiLSTM fusion model.
5. Accuracy/Metrics	AVEC 2016: 85.11% Accuracy, 78.79% F1. LOSOCV: 91.01% Accuracy, 85.95% F1.
6. Research Gaps	high GPT-4 inference time, Limited generalization; small dataset; Predicts only depression.



Article 9: Epidemiology of Common Mental Disorders (CMDs) in India — NMHS 2015–16 [14]

Written By	Pavithra Jayasankar, Narayana Manjunatha, Girish N. Rao, and NMHS India Collaborator Group
Based On	National Mental Health Survey (NMHS) India 2015–16 covering adults (18+) across India
Method Used	MINI v6.0.0 (ICD-10 based) assessment and Multiple Logistic Regression analysis
Key Findings	CMDs are common and often untreated in India. Higher risk was found among females, middle-aged, less educated, and unemployed individuals, with a significant treatment gap.

Article 10: Family Mental Health in India — A Systematic Review (2015–2025) [15]

Written By	Justin Raj, Nishant Goyal, and Senthil M.
Based On	Systematic review of 50 studies published between 2015–2025
Method Used	Literature review and synthesis of cross-sectional, longitudinal, intervention, qualitative, and meta-analysis studies
Key Findings	Family dynamics, structure, conflicts, parent–child relationships, caregiver burden, stigma, cultural beliefs, and gender bias significantly influence mental health outcomes in India.



Overall **Research Gaps** from Literature Study

Single-modal systems

Most existing models rely on only one type of data, limiting prediction accuracy.

No family analysis

The influence of family environment and relationships is rarely incorporated into mental health prediction.

No loneliness assessment

Loneliness, a major risk factor for mental health disorders, is often overlooked..

Black-box predictions

Many AI models provide predictions without explaining the reasons behind them.

Lack of Indian datasets

Most studies use foreign datasets that may not reflect Indian social and cultural conditions.

Innovative Solutions

Multimodal Framework

Integrate tabular, image, audio, and physiological data for comprehensive mental health assessment.

Explainable AI

Use XAI techniques to make model decisions transparent and understandable.

Family Condition Analysis

Incorporate family-related factors to evaluate their impact on mental health outcomes.

Loneliness Assessment

Include loneliness risk indicators as a key prediction in form of a mental disorder

Indian Synthetic Dataset

Develop an India-focused dataset that captures regional and cultural characteristics.



Problem Statement

“An **Explainable Multimodal AI** Framework for Predicting Mental Health and Analysing the Effect of Family Conditions and Loneliness.”

The background of the slide features a repeating pattern of stylized, four-lobed shapes (resembling rounded crosses or clovers) in shades of blue and green, arranged in a grid-like fashion.

Analysis Phase

Major Objective

"To develop an **Explainable Multimodal AI Framework** for predicting mental health conditions and studying the effects of family conditions and loneliness of the **Indian population** on mental health. The system will use Explainable AI to provide accurate, transparent, and easy-to-understand predictions for early mental health assessment."

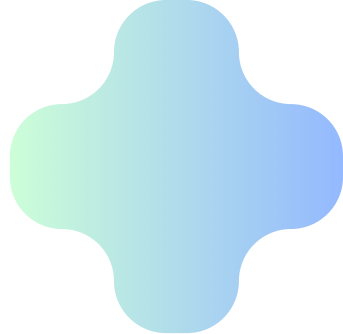
Sub-Objectives:

- **Objective 1:** Develop a multimodal AI framework using questionnaire (tabular) and audio data for mental health prediction.
- **Objective 2:** Improve the transparency and interpretability of predictions using Explainable AI (XAI).
- **Objective 3:** Predict mental health conditions while analyzing the impact of family factors and loneliness.
- **Objective 4:** Study mental health patterns in the Indian population using demographic, social, family, and cultural factors.



Functional Requirements

Sl No.	Functional Requirement	Input	Output	Description
1	User Registration and Authentication	User details, credentials	User account, authenticated session	The system shall allow users to securely register and access the platform.
2	Multimodal Data Collection	User responses	Structured assessment data	The system shall collect mental health, family, and loneliness-related information from users.
3	Real-Time Audio Capture and Analysis	User voice responses	Audio features (pitch, tone, speech rate, pauses, MFCCs)	The system shall capture and analyze audio responses during the assessment process.
4	Data Preprocessing and Feature Extraction	Questionnaire, audio, image, and heart rate data	Processed feature vectors	The system shall preprocess multimodal data for machine learning analysis.



Functional Requirements

5	Mental Health Prediction	Processed multimodal features	Predicted mental health condition and risk level	The system shall predict mental health conditions using AI models.
6	Explainable AI Analysis	Prediction results and model features	Explanation of influential factors	The system shall provide transparent explanations for prediction outcomes.
7	Family Influence Analysis, Recommendation and Report Generation	Prediction results, family-related data	Family influence score, recommendations, assessment report	The system shall analyze family influence, generate recommendations, and produce reports.
8	Assessment History and Data Retrieval	User request/history query	Historical assessment records	The system shall store and retrieve previous assessments for monitoring progress.

Non-Functional Requirements

Sl No.	Non-Functional Requirement	Description
1	Reliability	The system shall provide consistent and dependable results with minimal failures.
2	Performance	The system shall generate predictions, explanations, and reports within acceptable response times.
3	Scalability	The system shall support increasing numbers of users and assessments without significant performance degradation.
4	Usability / Human-Computer Interface	The system shall provide an intuitive, user-friendly, and accessible interface for all users.
5	Security	The system shall protect sensitive user information through secure authentication and data protection mechanisms.
6	Maintainability	The system shall be easy to modify, update, and maintain for future enhancements.
7	Privacy	The system shall ensure confidentiality of mental health, family, audio, image, and physiological data.

Technology Stack Requirements

Frontend



Next.js



React



TypeScript



Tailwind CSS

Backend



Python



FastAPI



JWT Auth



System Integration

Machine Learning & XAI



CatBoost



SHAP



Pandas



NumPy



Scikit-learn

Multimodal Data Processing



Librosa



Parselmouth



Questionnaire
Scoring

Database



PostgreSQL



Prisma ORM

Visualization, Reporting & Tools



Matplotlib



ReportLab



Git



GitHub

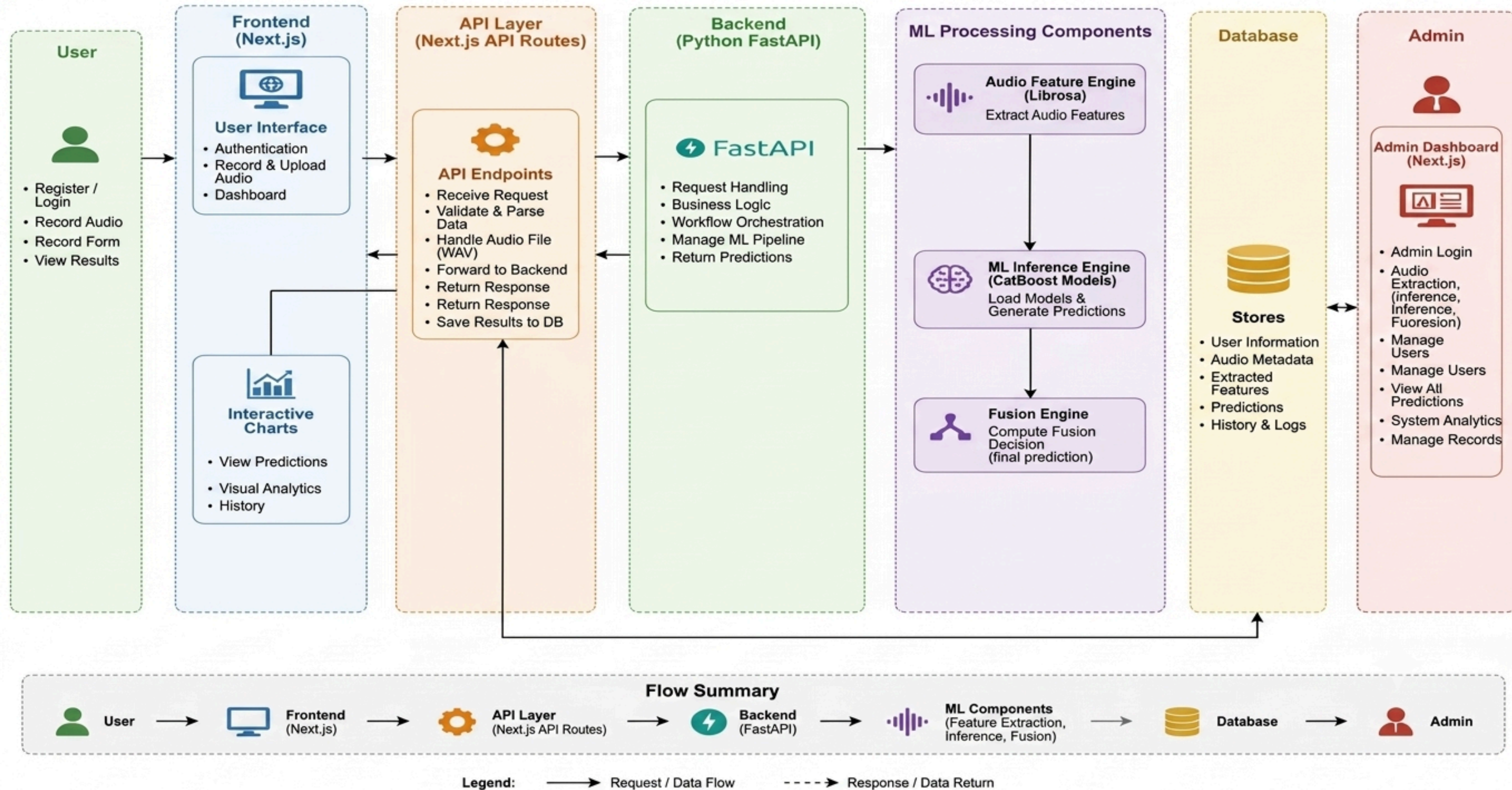


VS Code

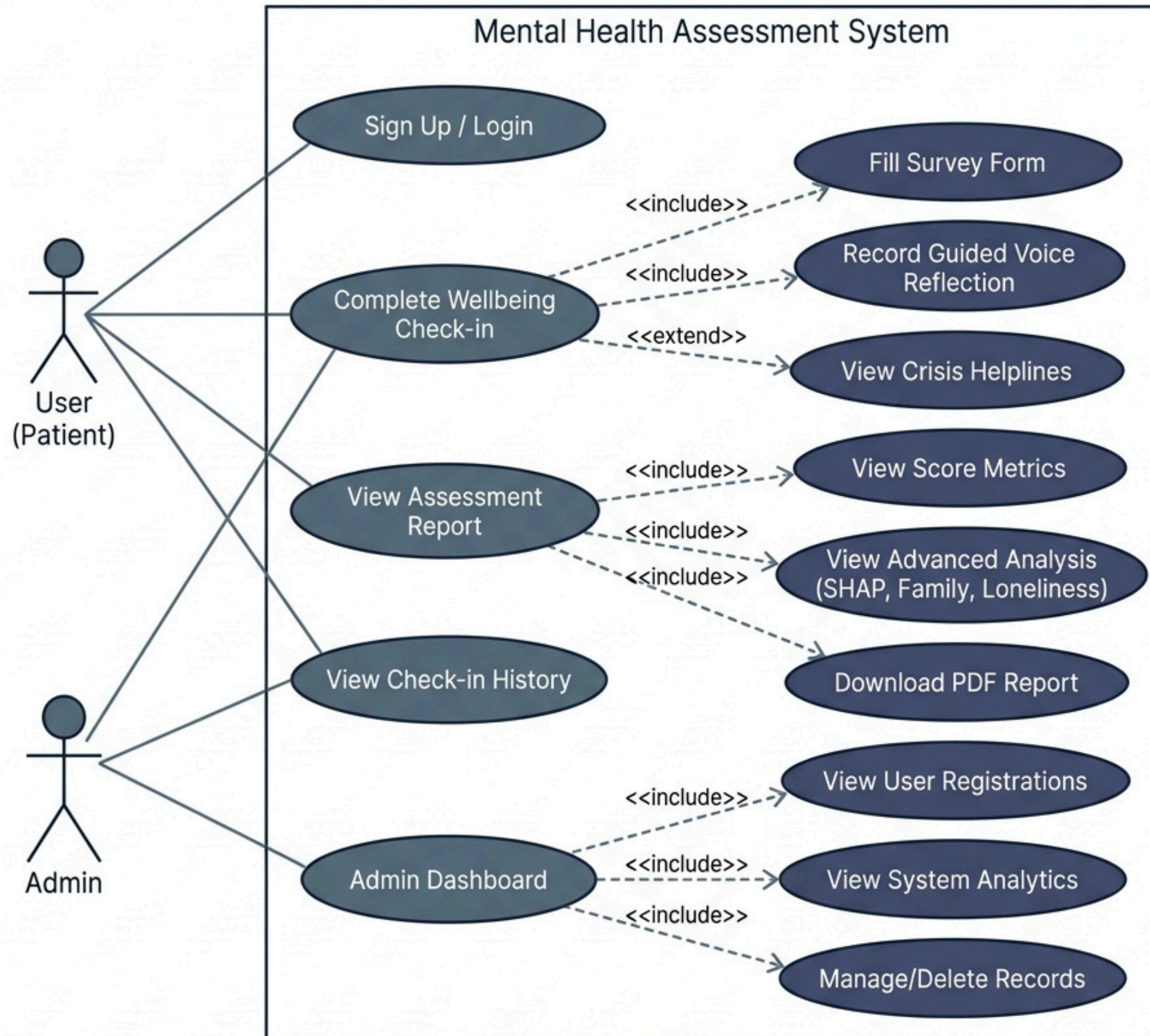


Design Phase

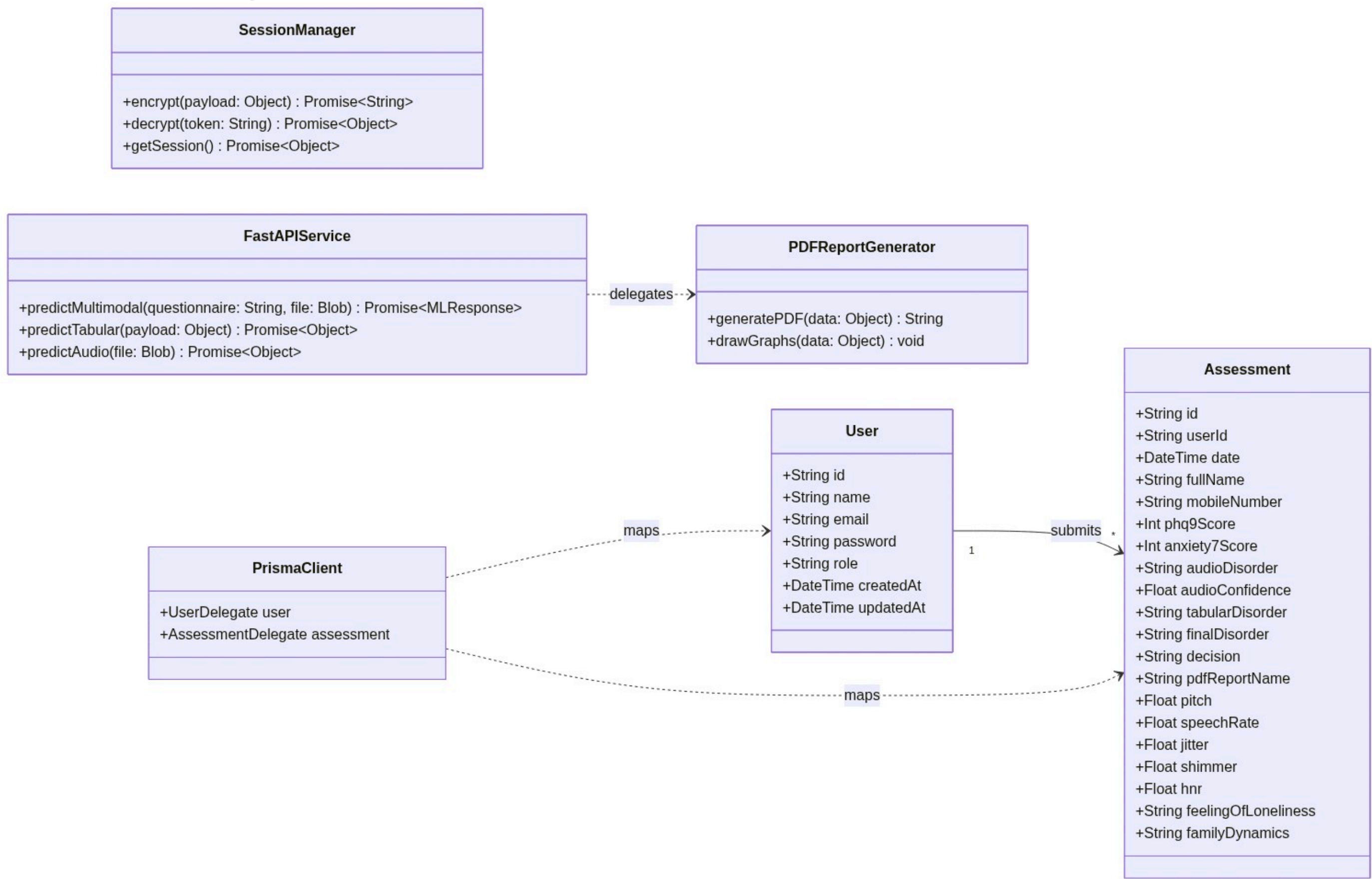
System Architecture



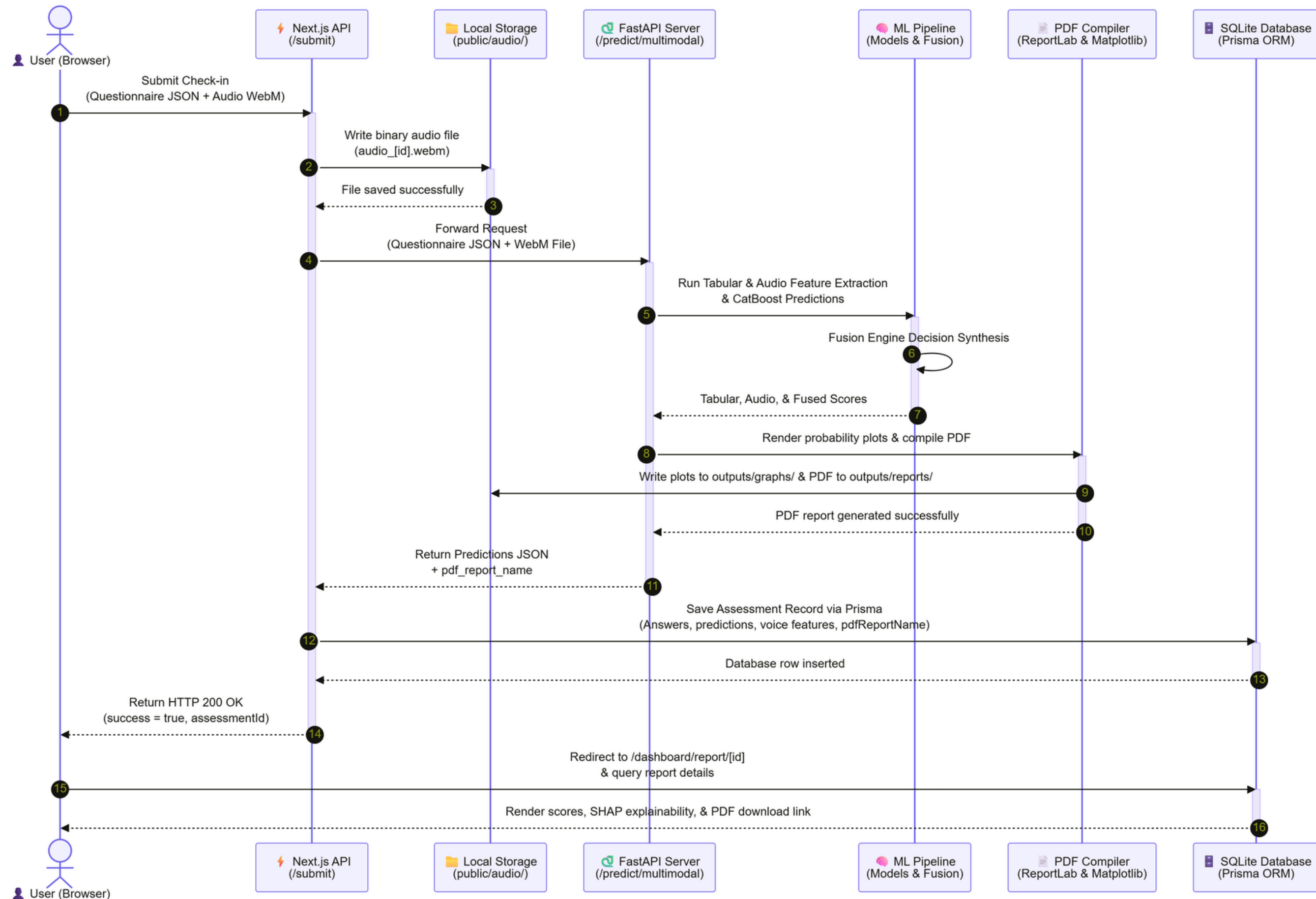
Use-case Diagram



Class Diagram

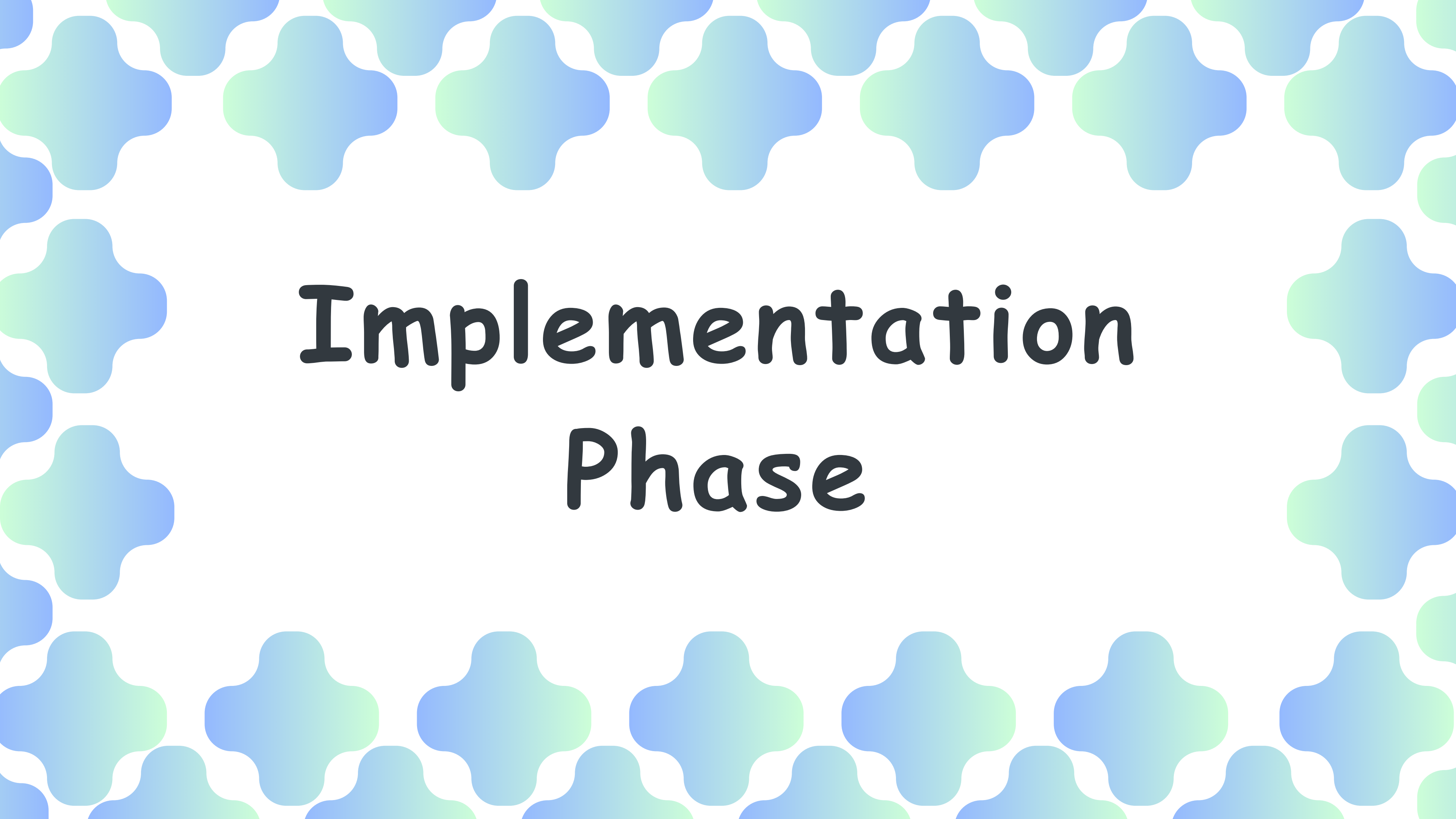


Sequence Diagram



Summary of planning, Analysis and Design phase

- **Planning:** Defined the system objectives, identified key stakeholders (User and Admin), and determined the core modules required for mental health assessment and prediction.
- **Analysis:** Analyzed user requirements and system interactions using the use-case diagram, specifying the workflow from assessment to prediction, reporting, and administration.
- **Design:** Designed the system architecture and class diagram, establishing the relationships between modules and the flow of data through collection, processing, AI prediction, explainability, and reporting.
- **Outcome:** Developed a scalable and modular system design that supports accurate mental health prediction, explainable AI insights, personalized recommendations, and secure report generation.



Implementation Phase



Methodology

Questionnaire of our project

Questionnaire Preparation and Mapping with Disorders:

1. Different diagnosis of major mental disorders are studied from **DSM 5 Book**[16] and 7 major disorder were selected: **Major Depressive Disorder(MDD)**, **Generalized Anxiety Disorder (GAD)**, **Post-Traumatic Stress Disorder (PTSD)**, **Persistent Depressive Disorder (Dysthymia)**, **Anxiety–Depression Comorbidity**, **Bipolar I**, **Bipolar II** and 4 risks: **Suicide Risk**, **Psychological Distress**, **Family-Related Mental Health Risk**, **Loneliness Risk**.
2. First common diagnosis of all 7 disorders and 4 risks were taken as attributes and also as questions for questionnaire and their possible values were categorized.
3. **PHQ 9** and **GAD 7** Questionnaire were also integrated with PHQ 2 and GAD 2 logic to extend our questionnaire.
4. Then the attributes were mapped directly and family related attributes were indirectly mapped for each disorder.

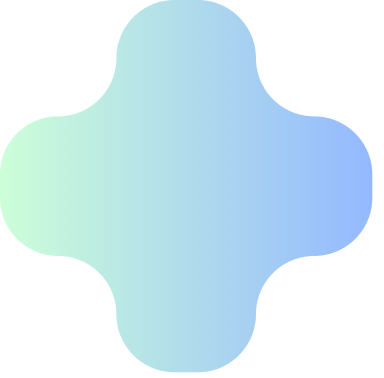
Ex - Considering **PTSD**

Diagnosis 1: Avoidance of or efforts to avoid distressing memories, thoughts, or feelings about or closely associated with the traumatic event(s).

Direct Mapping: Avoidance Behavior = Yes

Diagnosis 2: Feelings of detachment or estrangement from others.

Indirect Mapping: Emotional Support: Unavailable and Family Dynamics: Dysfunctional



Our Questionnaire

General Questions

- **Mood Changes:** Do you frequently experience sudden changes in mood?(y/n)
- **Optimism Level:** How optimistic do you feel about your future?(1-10)
- **Current Emotional State:** Which emotion best describes how you usually feel? (Anger/Fear/Happy/Neutral/Sad)
- **Overthinking:** Do you frequently overthink or repeatedly worry about situations?(Yes/Sometimes/No)
- **Social Fear:** How often do you feel nervous or fearful in social situations?(High / Moderate / Low)
- **Concentration:** How would you rate your ability to concentrate on daily activities?(1-10)
- **Social Interaction:** How would you describe your level of social interaction?(High / Moderate / Low)
- **Sleep Pattern:** How has your sleep been during the past month?(Normal / Reduced / Excessive)
- **Reduced Need for Sleep:** Do you sometimes feel energetic despite sleeping very little?(Yes/No)
- **Increased Energy Level:** Do you experience periods of unusually high energy or activity?(Yes/No)
- **Suicidal Thoughts:** Have you experienced thoughts of self-harm or suicide?(Yes/No)
- **PHQ-9 Assessment:** PHQ-9 Score:(0–27)
- **GAD-7 Assessment:** GAD-7 Score:(0–21)
- **Past Trauma:** Have you experienced a traumatic event that continues to affect you?(Yes/No)
- **Intrusive Memories:** Do you frequently experience unwanted memories or flashbacks of a traumatic event?(Yes/No)
- **Avoidance Behaviour:** Do you avoid people, places, or situations that remind you of a traumatic event? (Yes/No)

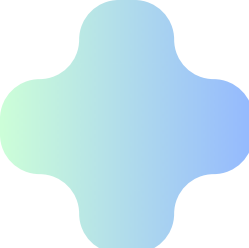
Our Questionnaire

Loneliness Assessment

- **Feeling of Loneliness:** How often do you feel lonely or isolated from others?
(Never/Rarely/Sometimes/Often/Always)
- **Feeling Understood:** Do you feel that people around you understand and support you?
(Always/Sometimes/Rarely/Never)

Family Environment Assessment

- **Family Structure:** What best describes your family structure?(Joint Family, Nuclear Family, Other (Orphan / Single Parent))
- **Family Dynamics:** How would you describe your family environment?
(Supportive/Neutral/Dysfunctional)
- **Marital and Family Conflict:** How often do conflicts occur within your family?(No Conflict/Occasional/Frequent)
- **Financial Stress:** What level of financial stress does your family experience?(High / Moderate / Low)
- **Emotional Support:** How much emotional support do you receive from family members?
(Available/Partially Available/Unavailable)



Synthetic Data Generation Method

Creating synthetic tabular dataset:

Non Weighted Dataset: This dataset was generated by directly giving the mapping of attributes and disorder without giving weight to any attribute.

Drawback: During Train-Test we found the overall accuracy was low and accuracy for each test record was even lower.

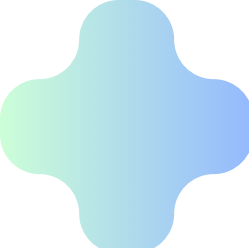
Weighted Dataset: This dataset was generated by giving weightage for all attributes based on their priority mentioned in DSM 5[16] .

Drawback: Perfect accuracy, as mental disorder symptom varies from person to person weighted data will not predict accurately sometime, no need of ML as using weight diagnosis can be done directly.

Hybrid Dataset: This data set was manually prepared from non weighted and weighted datasets combined in ratio of **3:2** which was decided by studying the ratio of mental health cases showing classical to non classical symptoms as per study of WHO.

Advantage: Good accuracy, similar to real world scenario, using of ML is possible.

This dataset have **4500 rows** approx which is sufficient to train the model.



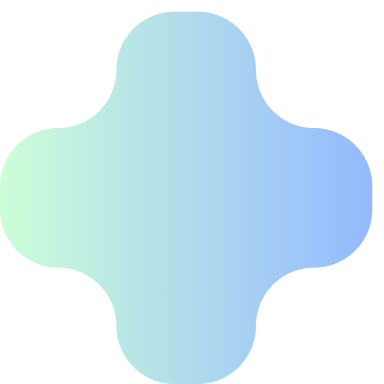
Creating synthetic audio features tabular dataset:

Audio Features: Identification of different audio features influencing different disorders from different research based study[17][18][19][20][21]. Following are identified features:

- Pitch(Hz)
- Pitch Variability(Hz)
- Speech Rate(WPM)
- Pause Duration(s)
- Voice Energy(RMS Amplitude)
- Jitter(%)
- Shimmer(%)
- HNR(Harmonic-to-Noise Ratio)(dB)

Categorical valued dataset: First we have generated the table with values like high, moderate, low etc for each features of the audio, but during execution in signal processing the value generated was in numeric value with unit.

Numerical valued dataset: Based on our requirement we have generated a numerical valued dataset for processing on the features extarcted by signal processing. It has approx 2500 rows.



Selection of Models

Selection for processing tabular dataset based on questionnaire
(by comparing their accuracy scores)

CatBoostRegressor

```
R2 : 0.71
MAE : 2.4213
MSE : 42.6534
RMSE : 6.531
Explained Variance : 0.7103
MAPE : 66.89 %
Accuracy ±5% : 90.82 %
Accuracy ±10% : 96.13 %
Per Person Accuracy : 90.82 %
```

XGBoostRegressor

```
Overall R2 Score : 0.6841
Mean Absolute Error : 2.6244
Mean Squared Error : 46.2022
Root Mean Squared Error : 6.7972
Explained Variance Score : 0.6853
Mean Absolute Percentage Error : 74.88%
Accuracy (±5%) : 88.95%
Accuracy (±10%) : 95.34%
Per Person Average Accuracy : 88.95%
```

RandomForestRegressor

```
Overall R2 Score : 0.6585
Mean Absolute Error : 2.6390
Mean Squared Error : 47.9354
Root Mean Squared Error : 6.9235
Explained Variance Score : 0.6614
Mean Absolute Percentage Error: 78.76%
Accuracy (±5%) : 88.91%
Accuracy (±10%) : 94.97%
Per Person Average Accuracy : 88.91%
```

Selection for processing audio feature dataset (by comparing their accuracy scores)

CatBoostClassifier

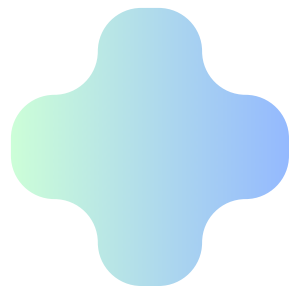
Accuracy	:	86.90%
Precision	:	86.91%
Recall	:	86.90%
F1 Score	:	86.89%

XGBoostClassifier

Accuracy	:	85.60%
Precision	:	85.61%
Recall	:	85.60%
F1 Score	:	85.58%

Signal Feature Processing(AudioData)

- **Librosa** is used to preprocess audio recordings and extract acoustic features for analysis.
- **Parselmouth (Praat)** is used to compute clinically relevant speech characteristics such as pitch, jitter, shimmer, and Harmonic-to-Noise Ratio (HNR).
- Extracts features including **pitch, pitch variability, speech rate, pause duration, voice energy, jitter, shimmer, and HNR**, converting speech into numerical feature vectors.
- The extracted features are provided to the CatBoost Classifier to verify questionnaire-based predictions before the final fusion decision.



Algorithms

Algorithm 1: Data Preprocessing and Model Training(Tabular.csv, AudioFeature.csv)

Description: This algorithm shows how the dataset are pre trained and the trained models were returned as “Disorder.pkl” file for each disorder. Tabular.csv is the questionnaire based dataset and AudioFeature.csv is dataset of audio features. This algorithm also returns Scalar.pkl(containing the mean and standard deviation calculated from the training data.) and Encoder.pkl(containing the mapping between categorical values and numerical values.)

1. Start
2. Load the tabular dataset.
3. Handle missing values and clean the data.
4. Encode categorical questionnaire features.
5. Separate input features and target variables.
6. Normalize numerical features where required.
7. Split the dataset into training and testing sets.
8. Train the CatBoost model for each disorder.
9. Save trained models as .pkl files.
10. Save encoders and preprocessing objects.
11. Load the audio feature dataset.
12. Separate audio features and target labels.
13. Normalize numerical audio features using StandardScaler.
14. Encode disorder labels using LabelEncoder.
15. Split the dataset into training and testing sets.
16. Train the CatBoost classifier.
17. Save the trained audio model as .pkl.
18. Save the scaler, label encoder, and feature names.
19. End

Algorithm 2: User Assessment and Multimodal Prediction(Questionnaire Responses, Audio Recording.wav)

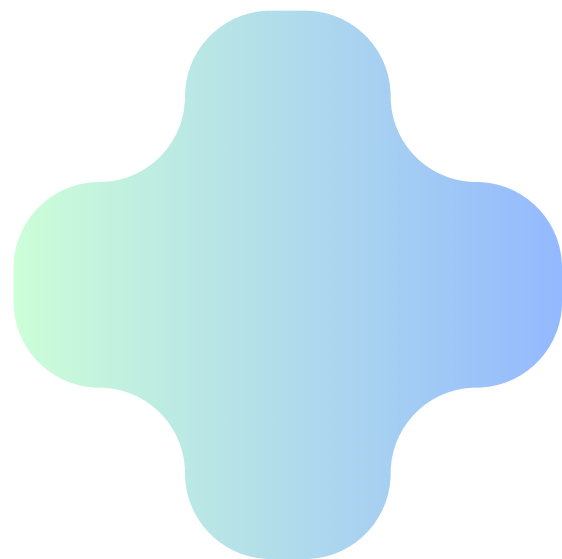
Description: This is the algorithm implemented for assessment of user. It takes user response and audio and gives Final Mental Health Diagnosis, Assessment Report, Assessment History.

1. Start
2. Collect questionnaire responses from the user.
3. Store questionnaire responses in questionnaire.csv.
4. Preprocess the questionnaire data using the saved encoder.
5. Load the trained tabular models (.pkl).
6. Predict disorder probabilities and identify the top three disorders.
7. Generate mental health risk flags.
8. Accept the user's audio recording (.wav).
9. Process the audio signal using Librosa and Parselmouth.
10. Extract acoustic features (Pitch, Pitch Variability, Speech Rate, Pause Duration, Voice Energy, Jitter, Shimmer and HNR).
11. Store the extracted features in audio_features.csv.
12. Normalize the extracted features using the saved scaler.
13. Load the trained audio classification model (.pkl).
14. Predict the audio-based mental health disorder and confidence score.
15. Fuse the questionnaire and audio predictions.
16. Determine the final mental health diagnosis.
17. Generate graphs and a PDF assessment report.
18. Store the assessment results in assessment_history.csv.
19. Delete the temporary input files (questionnaire.csv and audio.wav).
20. End

Algorithm 3: Admin Management Module(Admin Login Credentials, User Assessment Records, Training Datasets)

Description: This is the algorithm implemented for admin management. It takes admin log in information, user assessment data and gives Updated System Records, Retrained Machine Learning Models.

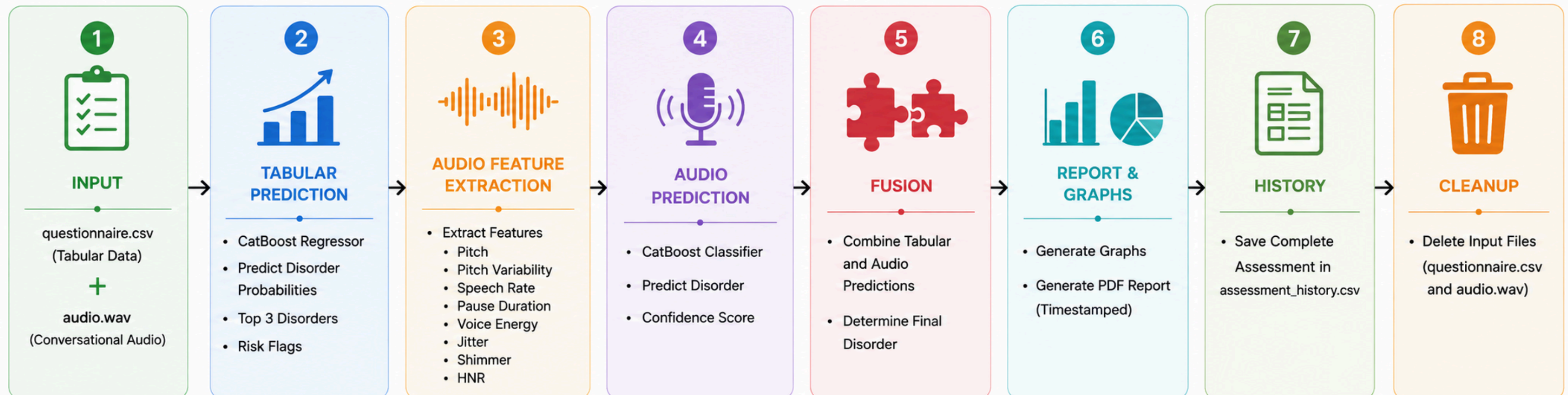
1. Start
2. Authenticate the administrator using valid login credentials.
3. Display the administrator dashboard.
4. View registered user information.
5. View assessment history and generated reports.
6. Monitor system analytics and prediction statistics.
7. Select assessment records for review or deletion.
8. Delete selected records from the assessment history.
9. Update the training datasets if required.
10. Retrain the questionnaire and audio machine learning models.
11. Save the updated trained models (.pkl) and preprocessing files.
12. Log all administrative actions.
13. Logout from the administrator dashboard.
14. End



Implementation Architecture

Modules Developed

- Questionnaire (Tabular) Prediction Module
- Audio Feature Extraction Module
- Audio Prediction Module
- Fusion Engine
- Report Generation Module
- Assessment History Module



Implementation

Machine Learning Pipeline

Questionnaire Model

- Data Preprocessing
- Feature Encoding
- Feature Scaling
- CatBoost Regressor
- Top 3 Disorder Prediction
- Risk Flag Generation

Audio Model

- Audio Loading
- Feature Extraction
- Feature Scaling
- CatBoost Classifier
- Disorder Prediction
- Confidence Score

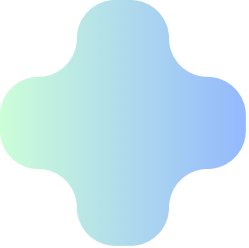
System Execution & Output

- **Automated Workflow:** The complete assessment is executed through a single file (main.py), eliminating the need to run individual modules manually.
- **Prediction Process:** The system processes questionnaire and audio inputs, predicts mental health disorders, and combines both results using the Fusion Engine.
- **Report Generation:** It automatically generates prediction graphs, a timestamped PDF report, and JSON files containing intermediate and final prediction results.
- **History & Cleanup:** The assessment is saved in the history database, and the input files are cleared automatically to prepare the system for the next assessment.



The background of the slide features a repeating pattern of stylized, four-lobed shapes (resembling rounded crosses or clovers) in shades of blue and green, arranged in a grid-like fashion around the central text.

Testing & Results



Testing & Validation

1. Unit & Functional Testing

API Endpoint Validation: Testing REST APIs (FastAPI backend) for multi-modal payload delivery (JSON survey responses + raw audio files).

Database Operations: Verification of schema migrations, user registrations, and admin management scripts (`promote.js`, `reset-password.js`) via Prisma.

Component Testing: Validating frontend components, including collapsible report summaries, visual range indicators (observed vs. clinical normal bounds), and tooltips.

2. Integration & End-to-End (E2E) Testing

Authentication & Redirection Flow: Testing JWT cookie-based middleware redirections (e.g., verifying that accounts with `ADMIN` privileges bypass delay syncs and land directly on `/admin`).

Web-to-PDF Parity: Ensuring the FastAPI ReportLab generator outputs PDF charts and data metrics that perfectly match the frontend dashboard styling.

Live API Sync: Verifying the cache-bypassing PDF download feature (`force=true`) to force real-time server recompilation upon user request.

Testing & Validation

3. Performance & Stability Optimization

Memory & Crash Resilience: Testing Matplotlib under headless mode (``Agg``) to ensure stability in containerized deployment environments (Render/Railway) without server crashes.

Cold-Start & Timeout Mitigation: Testing frontend request handling against backend cold-starts, optimizing API proxy timeout parameters to prevent Vercel 504 gateway timeouts.

Build System Verification: Resolving Windows-specific Turbopack watch-panics by falling back to a stable Webpack configuration.

Results & Discussion

Machine Learning Performance

- **Tabular Classifier (CatBoost):** Achieved high classification accuracy on the 4,500-row hybrid dataset trained using a 3:2 ratio of non-weighted and weighted DSM-5 symptom mappings.
- **Audio Feature Classifier:** Successfully validated on a 2,500-row numerical audio dataset containing speech rate, pitch, jitter, shimmer, and related acoustic features.

Explainable AI (XAI) Validation

- **Clinical Alignment:** SHAP explanations identified Loneliness and Family Dysfunctional Dynamics as major contributors to Depression and Anxiety risk, consistent with DSM-5 criteria.
- **Model Interpretability:** Generated feature-level explanations to improve transparency and support clinical understanding of predictions.

Results & Discussion

System Performance

- **Low Response Time:** End-to-end processing (audio upload → feature extraction → prediction → JSON response) completed in less than 3.5 seconds.
- **Efficient Resource Usage:** Optimized audio processing with an average RAM consumption of approximately 250 MB, ensuring stable system performance.

Report Generation Validation

- **PDF Integrity:** Successfully generated professional PDF reports without formatting or graphical distortions.
- **Visualization Consistency:** Maintained accurate rendering of dynamically generated Matplotlib charts within ReportLab-generated reports.



Conclusion & Future Scope

Roadmap of our project

Task / Activity	Week 1 (1–7 Days)	Week 2 (8–14 Days)	Week 3 (15–21 Days)	Week 4 (22–28 Days)	Week 5 (29–35 Days)	Week 6 (36–42 Days)
 Problem Statement Selection						
 Literature Study						
 Requirement Analysis						
 Design Analysis						
 Implementation						
 Testing						
 Deployment						
 Feedback Collection						
 Report Writing						
 Final Submission						

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Problem Statement Selection, Literature Study	Literature Study, Requirement Analysis, Design Analysis	Literature Study, Design Analysis, Implementation, Testing	Literature Study, Implementation, Testing	Implementation, Testing, Deployment, Feedback Collection	Report Writing, Final Submission

Conclusion and Future Work

- Developed an Explainable Multimodal AI Framework integrating questionnaire data and audio analysis for early mental health prediction, with XAI to identify key personal and family-related influencing factors.
- Designed and implemented a web-based application that enables user data collection, mental health prediction, explainable results, and personalized recommendations through an accessible interface.
- **Future Scope:**
 1. **Video Processing Integration:** Incorporate facial expression and behavioral analysis from video to improve mental health prediction accuracy.
 2. **Physiological Signal Analysis:** Integrate heart rate and other physiological signals using dedicated wearable devices for multimodal assessment.
 3. **Clinical Validation:** Validate the generated prediction reports through collaboration with mental health professionals and clinical studies.

Conclusion and Future Work

4. **Interactive Web Platform:** Develop a more user-friendly web application with advanced visualizations and enhanced user engagement features.
5. **Personalized Advice Module:** Provide personalized mental health recommendations, coping strategies, and wellness guidance based on prediction results.
6. **Screen Time & Content Analysis:** Analyze users' screen time and digital content consumption patterns to identify potential mental health risk factors.
7. **Expanded Multimodal Framework:** Combine questionnaire, audio, video, and physiological data for more comprehensive and reliable mental health assessment.
8. **Real-Time Monitoring:** Enable continuous monitoring and periodic reassessment to track changes in users' mental well-being over time.

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Thank You